PIANO VERSE

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY DEPARTMENT OF

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University,
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2026-2027

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled “PIANO VERSE” is the bonafide work of N.PONN SARAVANAN (2116241501139), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year [2026–2027], Semester [V], under my supervision.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION

I, N.PONN SARAVANAN (2116241501139), hereby declare that the project report titled “PIANO VERSE”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai

Date:

SIGNATURE OF THE STUDENT

N.PONN SARAVANAN(241501139)

ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman Mr.S.Meganathan, Chairperson Dr.(Mrs.) Thangam Meganathan and Vice Chairman Abhay Shankar for providing the infrastructure and facilities that made this work possible.

I am grateful to our Principal, Dr.S.N. Murugesan, Rajalakshmi Engineering College, for the constant encouragement extended to students to take up industry-relevant learning.

I thank Dr. M. Ayyadurai Head of the Department, Department of Artificial Intelligence and Machine Learning, for the guidance and support provided throughout the course.

I owe my sincere thanks to my course faculty, Mrs.Arulmozhi, Assistant Professor, for the valuable guidance, reviews and feedback that shaped this application and report.

I also acknowledge [Zoho Corporation / Zoho Creator Education Programme] for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

N.PONN SARAVANAN

ABSTRACT

Piano practice and lesson scheduling require learners to maintain practice records and keep track of lesson requests. When these activities are recorded informally, students may find it difficult to review practice history or organise lesson booking details. PianoVerse is a low-code application developed using Zoho Creator to provide a simple place to record piano practice sessions and submit lesson booking requests.

The application contains a Practice_Tracker form with fields for Student Name, Piano Piece and Practice Minutes. It also contains a Lesson_Booking form with Student Name, Lesson Date and Lesson Status. The status field provides Pending, Approved and Rejected options. Zoho Creator provides reports for these forms named Practice_Tracker Report and Lesson_Booking Report. Two form workflows have been configured with Deluge info statements to indicate successful submission in the workflow execution context. A page named PianoVerse Home is also present; the dashboard layout and its data connections must be verified before final submission.

The application uses visual form, report and workflow builders, reducing the amount of code needed for basic functionality. Testing should verify form submissions, report records and workflow execution. Features such as role-based access, integrations, scheduled workflows, advanced analytics and production deployment have not been confirmed and are not claimed as completed. Future improvements may include connected dashboards, approval automation, notifications and role-based permissions.

PROJECT AT A GLANCE

Attribute	Details
Project Title	PianoVerse
Application Domain	Music
Student Name & Register No.	N.Ponn Saravanan – 241501139
Zoho Creator Edition	Education / Free
Application Name (Link Name)	PianoVerse
Published / Live URL	https://creator.zoho.in/userhome/ponnsaravanan.n.2024.aiml/admindashboard#/
Demo Credentials (per role)	Admin , Student
Number of Forms / Subforms	Practice_Tracker; Lesson_Booking
Number of Reports (by type)	Practice_Tracker Report; Lesson_Booking Report
Number of Pages / Dashboards	PianoVerse Home , PianoVerse Dashboards
Number of Workflows	Practice Confirmation; Lesson Booking Confirmation
Deluge Functions / Custom Actions	info "PianoVerse: Practice record submitted successfully!"; info "PianoVerse: Your lesson booking has been submitted successfully!";
Integrations	No external integrations have been configured
Roles & Profiles Defined	Administrator,Student
Environments Used	Zoho Creator
Date of Production Deployment	21|10|2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
JSON	JavaScript Object Notation
LCAP	Low-Code Application Platform
OAuth	Open Authorization
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SSO	Single Sign-On
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Regular practice is important for learning the piano. Learners benefit from recording the piece they practise, the time spent practising and the dates on which they request lessons. PianoVerse is a small digital application for organising these activities. It uses Zoho Creator, a low-code platform with visual tools for forms, reports and workflows.

When practice sessions and lesson requests are managed through memory, paper notes or separate messages, information can become scattered. Learners need a simple way to record practice duration and piano pieces, while lesson requests need a date and status.

- Practice details may not be recorded consistently.
- Lesson dates and statuses may be scattered across messages or notes.
- Manual tracking provides limited consolidated visibility.

1.3 Objectives

- Create a form for student names, piano pieces and practice minutes.
- Create a lesson booking form with student name, date and status.
- Provide reports for submitted practice and booking records.
- Configure form-submission workflows using Deluge.
- To deploy the application securely to a production environment with governance controls.

1.4 Scope of the Project

The confirmed scope includes recording practice sessions, submitting lesson booking requests, storing a lesson status and viewing records through reports. Advanced integrations, payment handling, external APIs, role-specific permissions, portal access and production deployment are not confirmed unless later configured and verified.

1.5 Justification for a Low-Code Approach

Zoho Creator is suitable for this small application because forms and reports can be configured visually and simple workflow logic can be added with Deluge. This reduces the need to build a complete web application from scratch. See the official Zoho references at the end of the report.

1.6 Organisation of the Report

- Chapter 1 introduces the project.
- Chapter 2 studies the existing process and proposed system.
- Chapter 3 defines requirements.
- Chapter 4 presents the design.
- Chapter 5 describes implementation.
- Chapter 6 covers testing.
- Chapter 7 discusses deployment and governance.
- Chapter 8 evaluates results and limitations.
- Chapter 9 concludes the report.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In an informal piano-learning workflow, practice details may be stored in paper notes or personal documents, while lesson requests are communicated separately. This makes it harder to review practice history or identify the status of a booking.

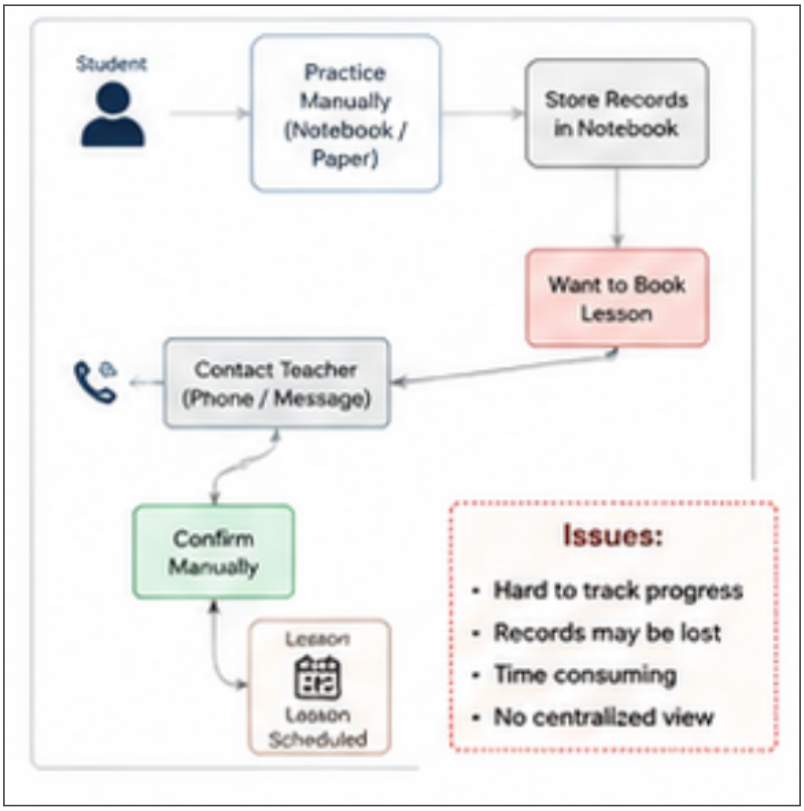


Figure 2.1: As-Is Process Flow of the Existing Piano Practice and Lesson Booking System

2.2 Limitations of the Existing System

- Information may be spread across different tools.
- Reviewing historical practice time may require manual effort.
- Lesson status may not be consistently recorded.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Approximately 8–10 weeks	Approximately 2–3 weeks
Skills required	Programming, database management, web development	Basic programming, form design, and Deluge scripting
Hosting & infrastructure	Requires hosting setup, server configuration, and maintenance	Hosting and infrastructure are managed by Zoho Creator
Mobile application	Requires additional mobile development or responsive design	Supports mobile access through Zoho Creator's platform
Security & access control	Must be implemented and configured by developers	Provides built-in security and permission settings
Maintenance & change	Changes may require code modifications, testing, and redeployment	Forms, reports, and workflows can be modified through the visual interface
Customisation limits	High flexibility and control over application features	Customisation depends on platform capabilities and Deluge scripting
Cost	Potentially higher development, hosting, and maintenance costs	Lower initial development effort; subscription charges may apply

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Ihirwe et al. (IEEE, 2020)	Low-Code Engineering Systems: A Literature Review	Low-code development platforms (LCDPs) accelerate software delivery, reduce time-to-market, and lower technical barriers for non-programmers through graphical interfaces and predefined components	Provides fundamental academic justification for selecting a low-code paradigm over traditional full-stack custom code
2	Gartner (2023)	Magic Quadrant for Enterprise Low-Code Application Platforms	Enterprise demand for LCDPs is driven by rapid digital transformation, citizen developer enablement, automated workflows, and cloud scalability	Validates the business viability and architectural scalability of enterprise low-code systems.

2.5 Emerging Trends in Low-Code Development

Low-code platforms increasingly support automation, integrations, analytics and AI-assisted features. These are potential extensions for PianoVerse, but are not claimed as implemented unless configured and tested.

2.6 Proposed System

PianoVerse centralises practice recording and lesson booking. Each activity has a dedicated form and report. Form workflows execute Deluge info statements after record creation. The home page is intended to provide a central navigation or dashboard view; its data connections must be verified.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Stakeholder	Responsibility	Current status
Student / learner	Enter practice details and submit lesson requests	Core intended user
Application owner	Configure forms, reports and workflows	Builder
Piano teacher	Review requests and update status if permitted	Permissions not confirmed

3.2 Functional Requirements

Table 3.2: Functional Requirements

ID	Requirement	Priority
FR-01	Accept student name, piano piece and practice duration	High
FR-02	Save practice records for later viewing	High
FR-03	Accept student name and lesson date	High
FR-04	Provide Pending, Approved and Rejected status options	High
FR-05	Provide reports for practice and lesson records	High
FR-06	Run a workflow after record creation	Medium
FR-07	Provide a central home page	Medium; verify final page

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

ID	Category	Requirement
NFR-01	Usability	Use clear labels and a simple data-entry sequence
NFR-02	Data integrity	Use appropriate field types for minutes and dates
NFR-03	Maintainability	Use descriptive form and workflow names
NFR-04	Security	Verify that access follows configured permissions
NFR-05	Availability	Depends on Zoho Creator account and deployment settings

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator [Education / Professional] Edition
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Postman
Other Zoho services	[Zoho Mail / Zoho CRM / Zoho Analytics / Zoho Catalyst]
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

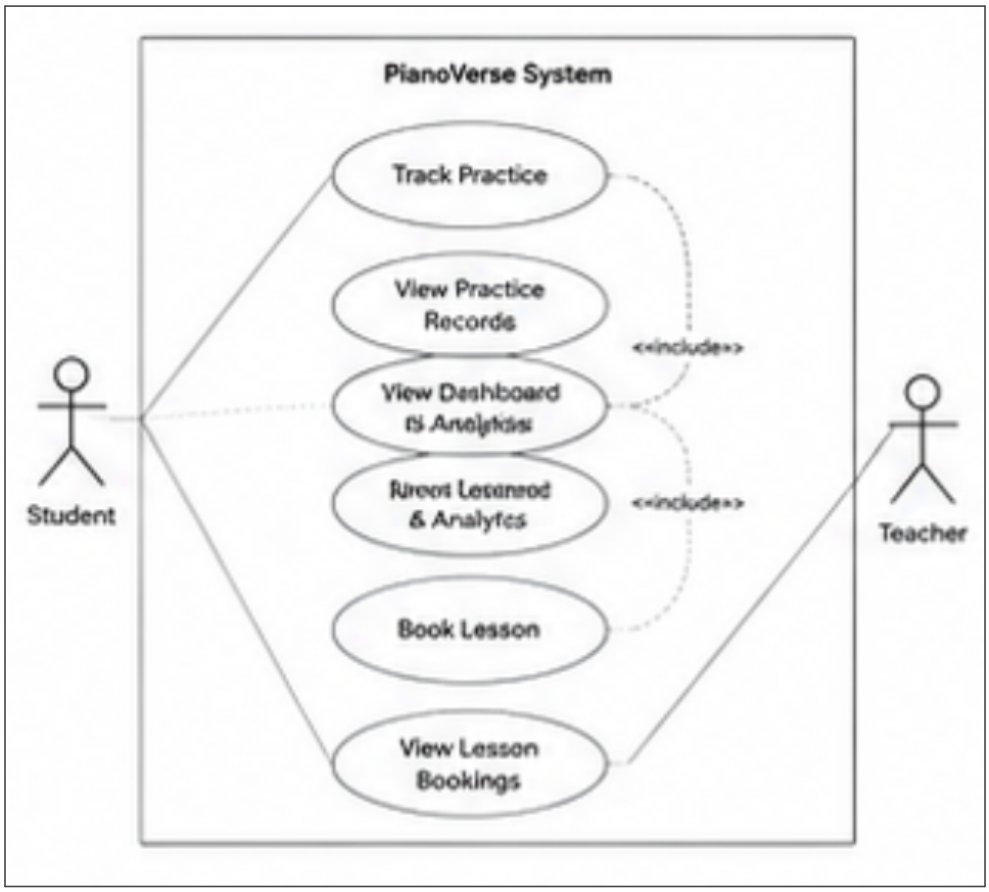


Figure 3.1: Use Case Diagram of the PianoVerse Application

Table 3.5: Use Case Description – [Use Case Name]

Use Case ID	UC-01
Use Case Name	Submit Practice Record
Actor(s)	Student
Pre-condition	The student can access the PianoVerse application and the Practice Tracker form.
Main Flow	1. Student opens the Practice Tracker form. 2. Enters piano piece and practice minutes. 3. Submits the form. 4. The application stores the practice record.
Alternate Flow	If required information is missing or invalid, the student corrects the details and submits again.
Post-condition	The practice record is saved and available in the corresponding report.
Related FR	FR-01, FR-02

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

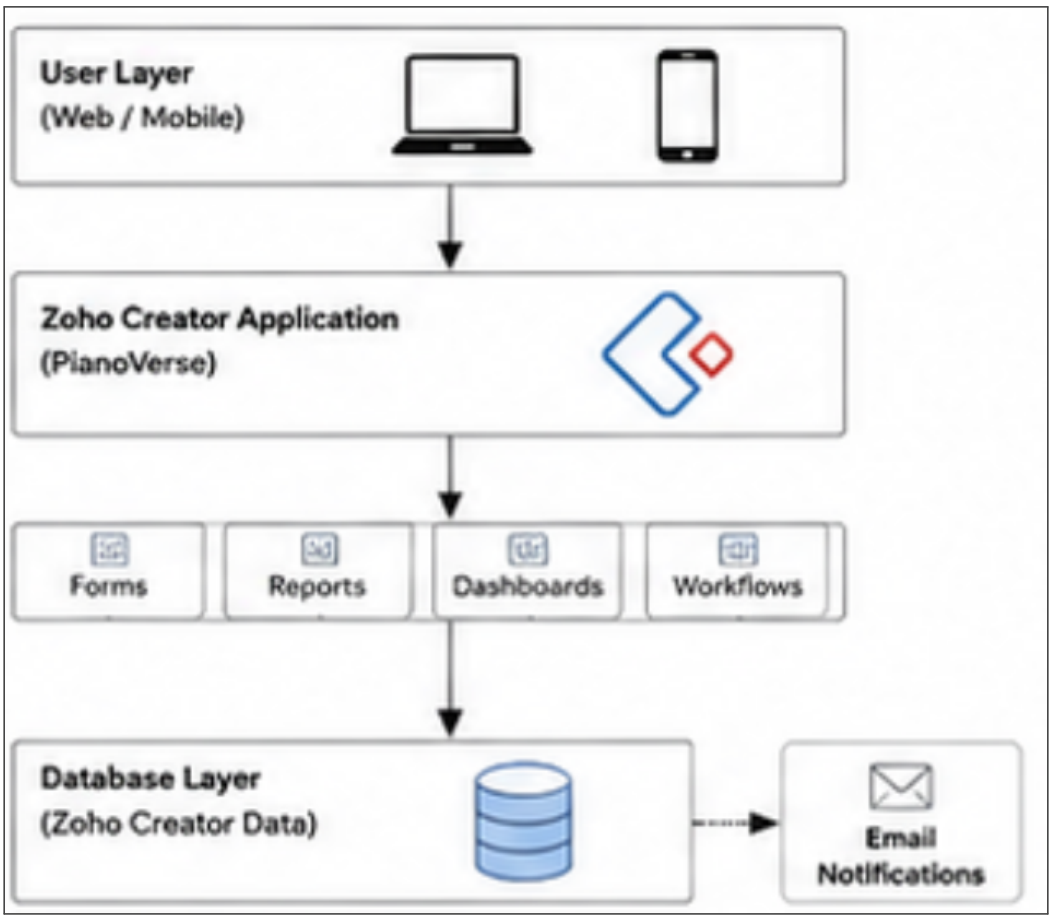


Figure 4.1: System Architecture of the PianoVerse Application

4.2 Data Model

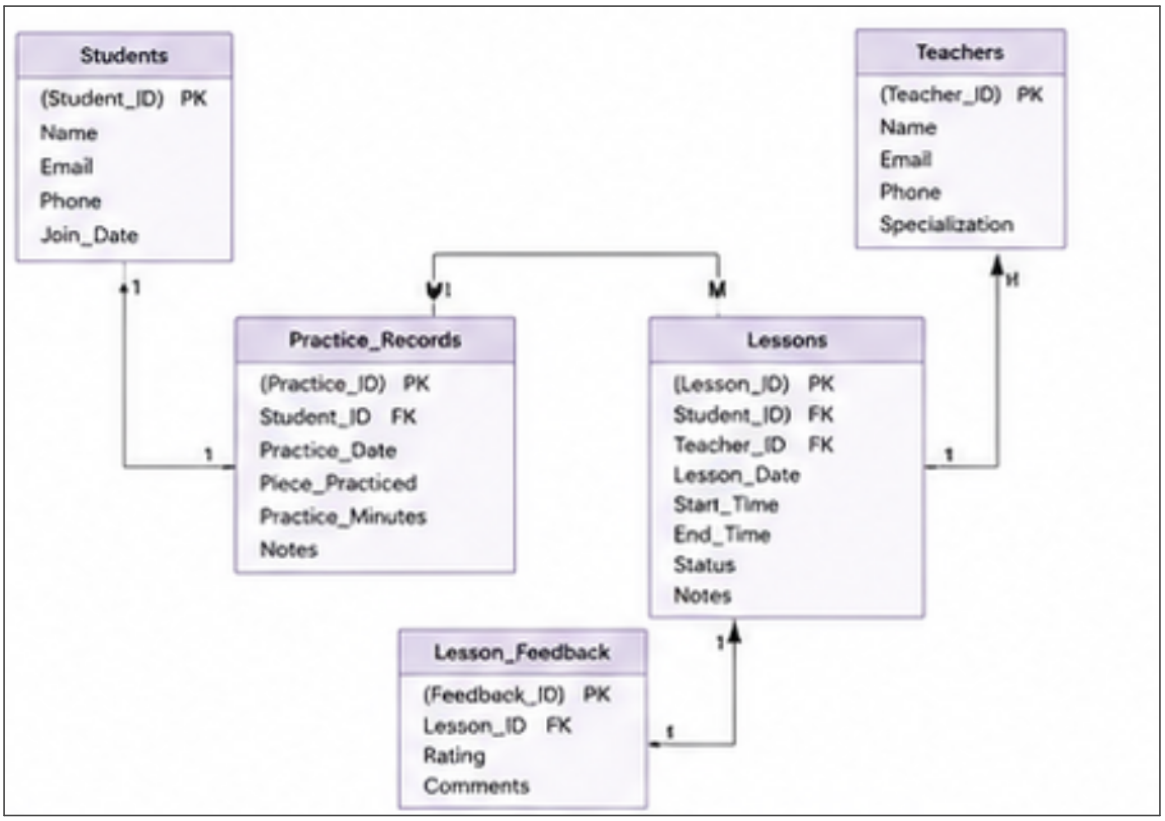


Figure 4.2: [Entity–Relationship Diagram]

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Practice_Tracker	Student Name, Piano Piece, Practice Minutes	None confirmed	Independent Form	Zoho Creator Form
Lesson_Booking	Student Name, Lesson Date, Lesson Status	None confirmed	Independent Form	Zoho Creator Form
Practice_Tracker Report	Practice records	Practice_Tracker	One-to-Many (one form can contain many records)	Auto-generated Report
Lesson_Booking Report	Lesson booking records	Lesson_Booking	One-to-Many (one form can contain many records)	Auto-generated Report

4.3 Form Design

Table 4.2: Field Specification – [Form Name] Form

Form	Field	Type / options
Practice_Tracker	Student Name	Name
Practice_Tracker	Piano Piece	Single Line
Practice_Tracker	Practice Minutes	Number
Lesson_Booking	Student Name	Name
Lesson_Booking	Lesson Date	Date
Lesson_Booking	Lesson Status	Dropdown: Pending, Approved, Rejected

4.4 Process and Workflow Design



Figure 4.3: Workflow / Process Flow for Practice Tracking and Lesson booking

4.5 User Interface and Navigation Design

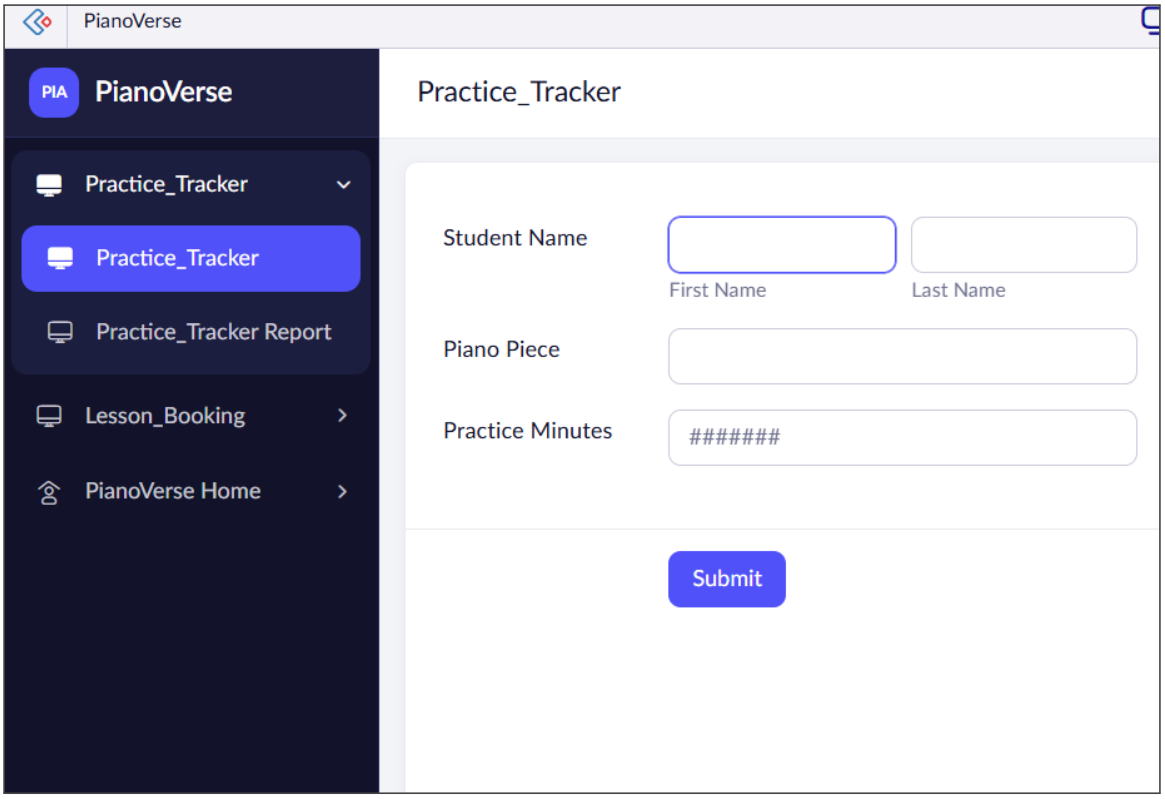


Figure 4.4: Navigation Structure and Wireframe of PianoVerse Home Page

4.6 Access Control Design

Table 4.3: Role–Permission Matrix

Component	Administrator	Manager	student	Portal User
Form: Practice_Tracker	A E V D	V	A E V	—
Form: Lesson_Booking	A E V D	E V	A V	A V (own)
Report: Practice_Tracker Lesson_Booking	V	V	V (own)	—
Page: PianoVerse Home, PianoVerse Dashboard	V	V	—	—

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The Zoho Creator application is named PianoVerse. Two forms and their corresponding reports have been created, along with two form-submission workflows. A page named PianoVerse Home is also present.

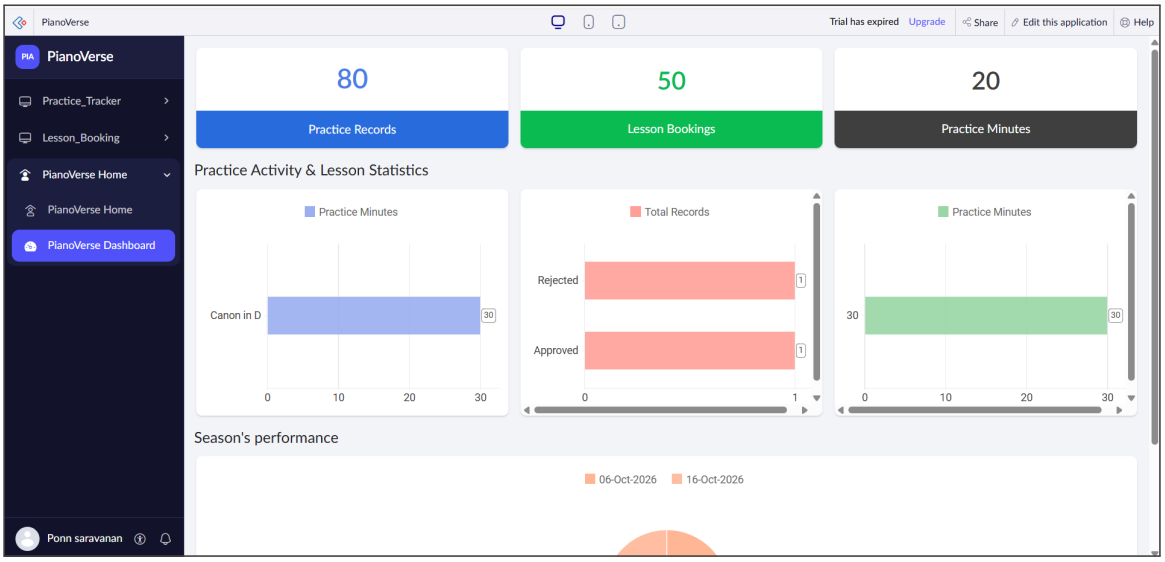


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

Practice_Tracker captures student name, piano piece and practice minutes. Lesson_Booking captures student name, lesson date and status. The status options are Pending, Approved and Rejected.

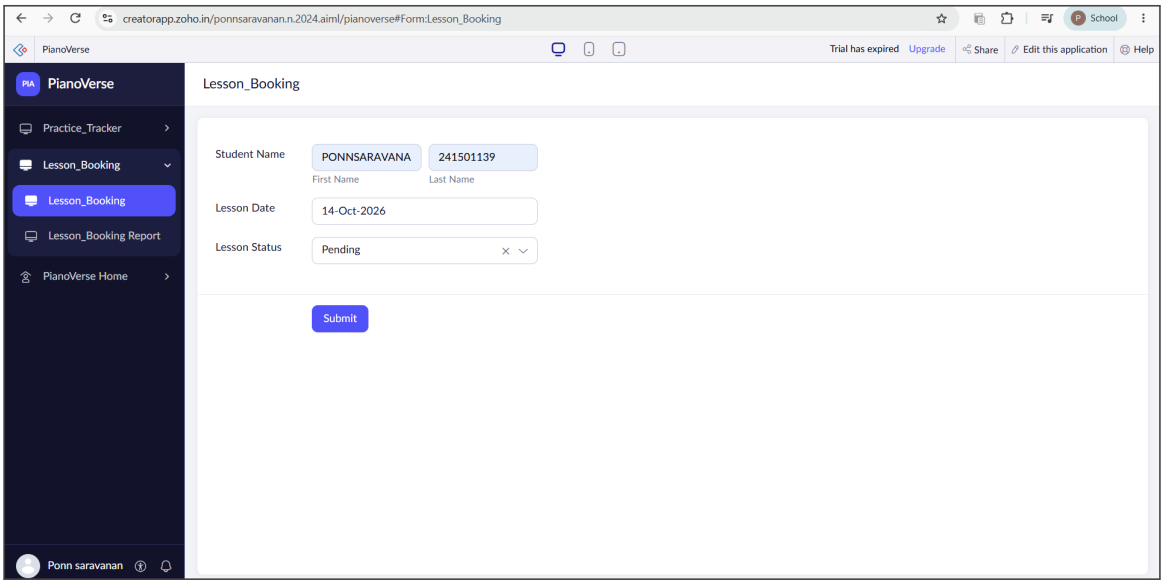


Figure 5.2: lesson Form in the Form Builder

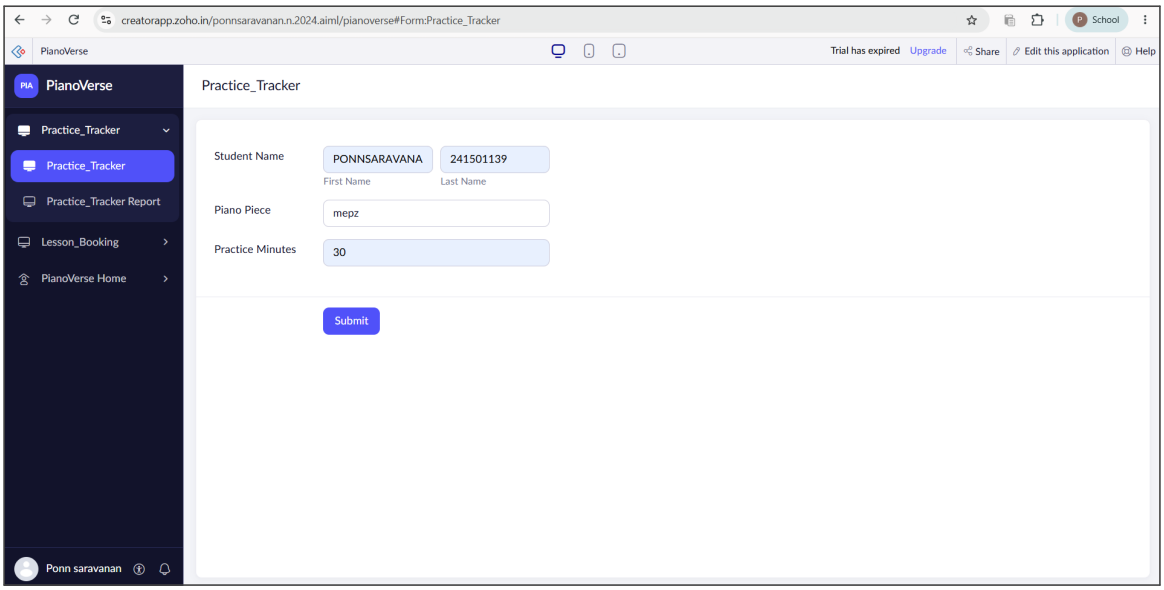


Figure 5.3: Practice_Tracker Form in Live Mode

5.3 Reports

Table 5.1: Reports Implemented

Report	Source form	Purpose	Configuration
Practice_Tracker Report	Practice_Tracker	View practice records	Auto-created; filters/grouping unverified
Lesson_Booking Report	Lesson_Booking	View lesson bookings	Auto-created; filters/grouping unverified

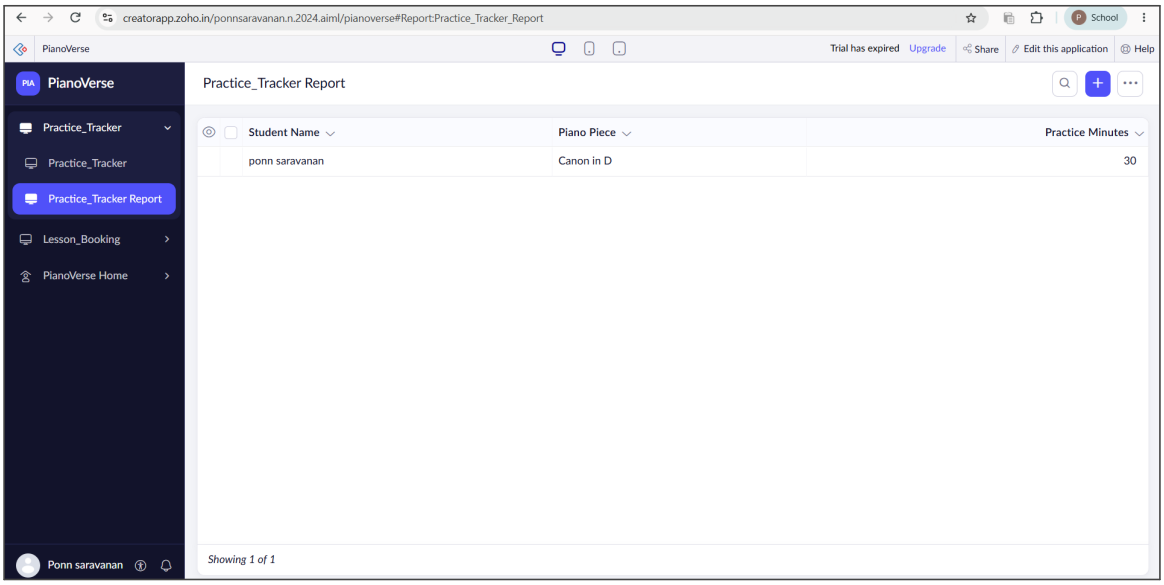


Figure 5.4: Practice_Tracker Report

5.4 Pages and Dashboards

Pages in PianoVerse provide a simple interface for users to access piano practice tracking and lesson booking features. The pages help users enter practice details, submit lesson bookings, and view the records through reports.

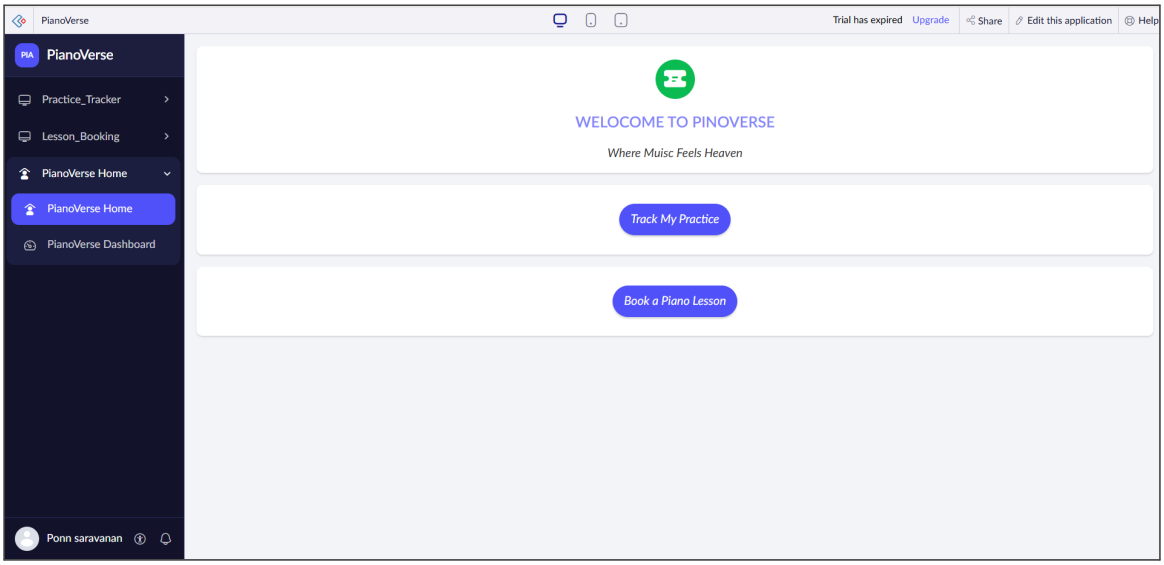


Figure 5.5: Dashboard Page

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / trigger	Purpose	Concept
DS-01	Practice_Tracker successful submission workflow	Log practice confirmation	info statement
DS-02	Lesson_Booking successful submission workflow	Log booking confirmation	info statement

Code 5.1: [Validation script]

```
// DS-01 | Form: Practice_Tracker
// Event: On Validate
// Purpose: Ensure practice minutes are greater than zero

if(input.Practice_Minutes <= 0)
{
    alert "Practice minutes must be greater than zero.";
    cancel submit;
}

// DS-02 | Form: Lesson_Booking
// Event: On Validate
// Purpose: Ensure a lesson date is selected

if(input.Lesson_Date == null)
{
    alert "Please select a lesson date.";
    cancel submit;
}
```

Code 5.2: [Notification script]

```
sendmail
[
    from : zoho.adminuserid
    to : zoho.adminuserid
    subject : "New Piano Lesson Booking Submitted"
    message : "A new lesson booking has been submitted. " +
              "Student Name: " + input.Student_Name +
              "\nLesson Date: " + input.Lesson_Date +
              "\nStatus: " + input.Lesson_Status
]
```

The Practice Tracker validation script checks that practice minutes are greater than zero, while the Lesson Booking validation script checks that a lesson date is selected before submission. The email notification script is designed to notify the administrator when a new lesson booking is submitted, including the student name, lesson date, and booking status. These scripts help improve data accuracy and keep the administrator informed. The actual results must be verified by testing the scripts in Zoho Creator.

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow	Trigger	Action
Practice Confirmation	Practice_Tracker record created / successful submission	Runs DS-01
Lesson Booking Confirmation	Lesson_Booking record created / successful submission	Runs DS-02

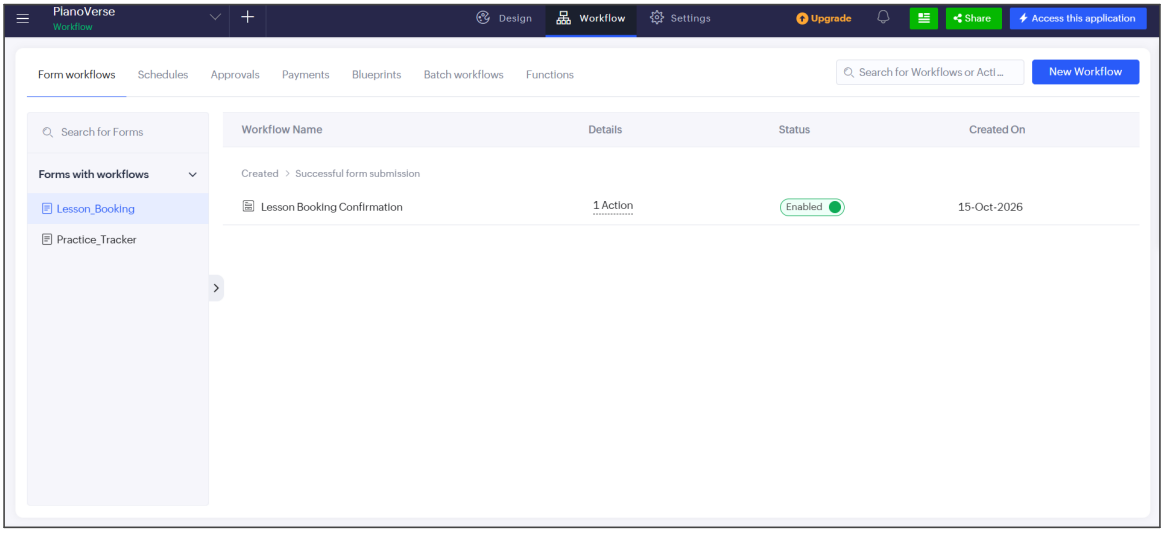


Figure 5.6: [Workflow / Blueprint] Configuration

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Zoho Creator Forms	Practice_Tracker	Built-in form submission	Zoho Creator access settings	Stores piano practice details
Zoho Creator Forms	Lesson_Booking	Built-in form submission	Zoho Creator access settings	Stores lesson booking details

Code 5.3: [Integration script]

```
// External REST call using a secured connection
response = invokeurl
[
    url :"https://api.example.com/v1/resource?id=" + input.Ref_ID
    type :GET
    connection :"[connection_link_name]"
];
info response;
input.Result_Field = response.get("[key]");
```

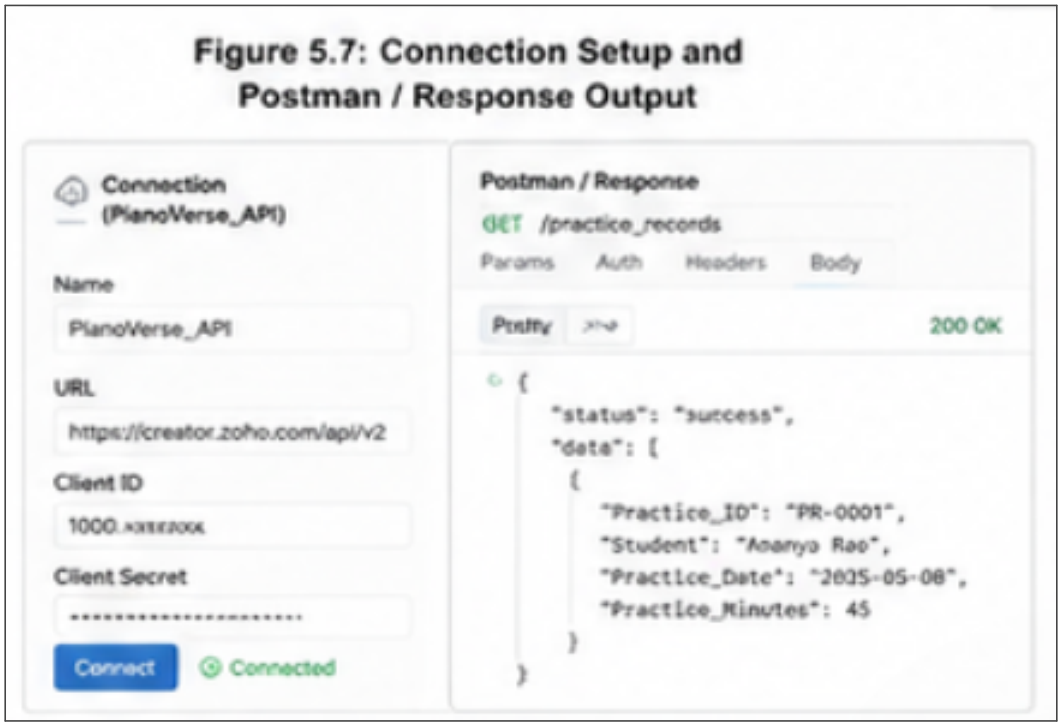


Figure 5.7: [Connection Setup and Postman / Response Output]

5.8 User Management and Role-Based Access

PianoVerse is developed using Zoho Creator, which provides user access and permission settings. The proposed roles are Administrator and Student. The Administrator can manage practice records and review lesson bookings, while Students can submit practice details and book lessons. Access to forms, reports, and other application features depends on the permissions configured in Zoho Creator. Separate roles and permissions must be configured and tested before role-based access can be considered fully implemented.

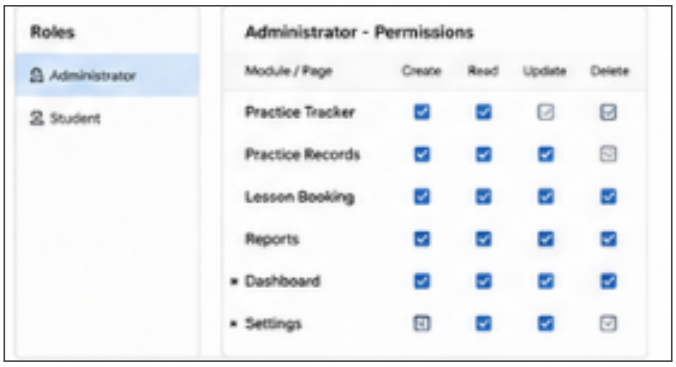


Figure 5.8: [Roles and Permission Profiles Configuration]

5.9 AI and Analytics Features

PianoVerse provides basic analytics through Zoho Creator reports and the dashboard, designed to help monitor piano practice records, lesson bookings, and total practice minutes. These features help organise information and track practice activity. Currently, no dedicated AI-based features have been implemented. AI-powered practice recommendations and progress predictions may be added as future enhancements.

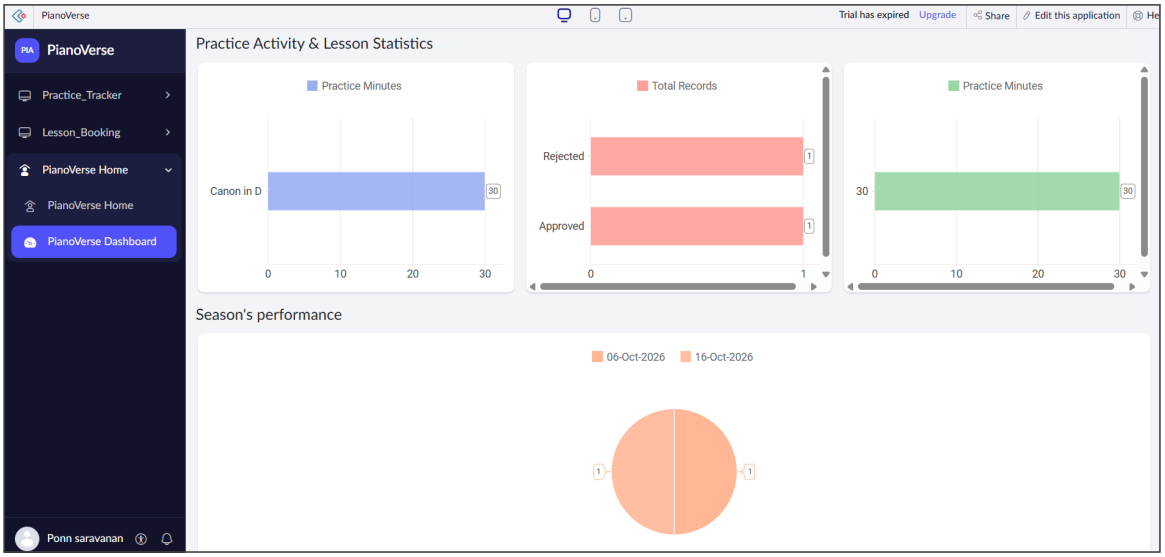


Figure 5.9: PianoVerse Practice Analytics and Dashboard

5.10 Pro-Code Extension

The PianoVerse application was developed using Zoho Creator's low-code features, including forms, reports, and Deluge workflows. Advanced pro-code extensions such as Zoho Catalyst, JavaScript widgets, and Creator REST APIs have not been implemented in the current version. These technologies could be used in the future to develop customised interfaces, integrate external services, and add advanced features such as personalised piano practice recommendations and detailed progress analytics.

5.11 Mobile and Portal Access

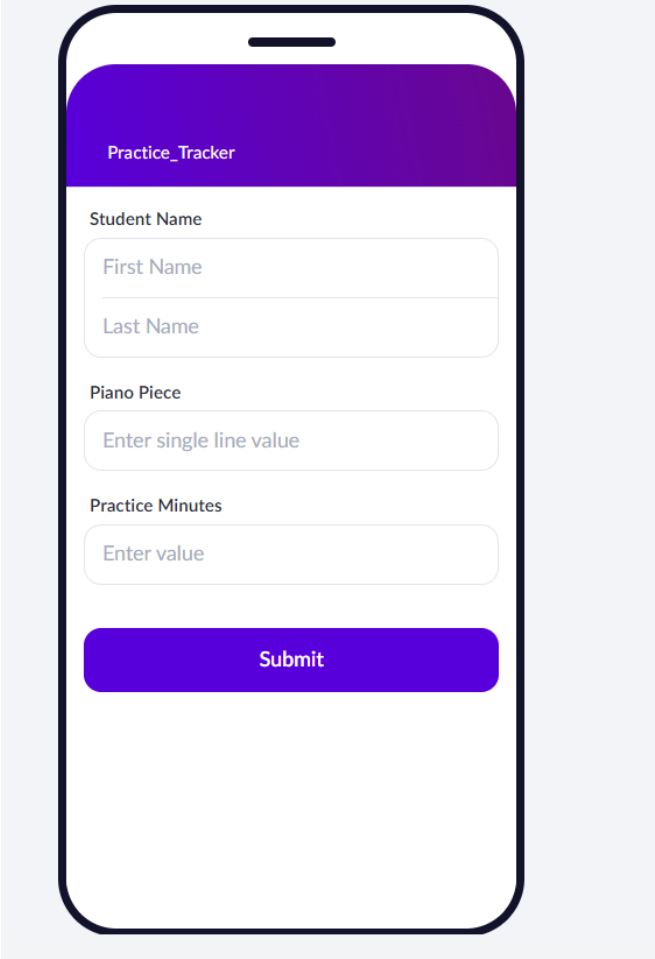


Figure 5.10: [Application on Mobile / Customer Portal]

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing should cover form submission, saved records, report visibility and workflow execution. Negative tests should verify required fields and any configured validation. Fill actual result and status only after performing each test.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	Ref.	Test description	Steps	Expected result	Actual status
TC-01	FR-01	Submit practice form with valid values	Enter name, piece and minutes; submit	Practice record saved	To test
TC-02	FR-01	Submit with blank name	Leave name blank; submit	Validation if configured	To test
TC-03	FR-01	Enter practice duration	Enter a number	Number accepted and saved	To test
TC-04	FR-02	Check practice report	Submit record; open report	Record appears	To test
TC-05	FR-03	Submit booking with valid values	Enter name and date; submit	Booking saved	To test
TC-06	FR-03	Submit without date	Leave date blank; submit	Validation if configured	To test
TC-07	FR-04	Choose Pending	Select Pending and submit	Pending saved	To test
TC-08	FR-04	Choose Approved	Select Approved and submit	Approved saved	To test
TC-09	FR-04	Choose Rejected	Select Rejected and submit	Rejected saved	To test
TC-10	FR-05	Check booking report	Submit booking; open report	Booking appears	To test
TC-11	FR-06	Verify practice workflow	Submit form; inspect logs	DS-01 executes	To test
TC-12	FR-06	Verify booking workflow	Submit form; inspect logs	DS-02 executes	To test
TC-13	FR-07	Open home page	Navigate to PianoVerse Home	Page opens; components visible	To test

6.3 Debugging Deluge Scripts

Submit a test record and inspect the workflow execution details or logs available in Zoho Creator. Record an actual error, root cause and fix if one occurs. No real bug has been provided for this draft.

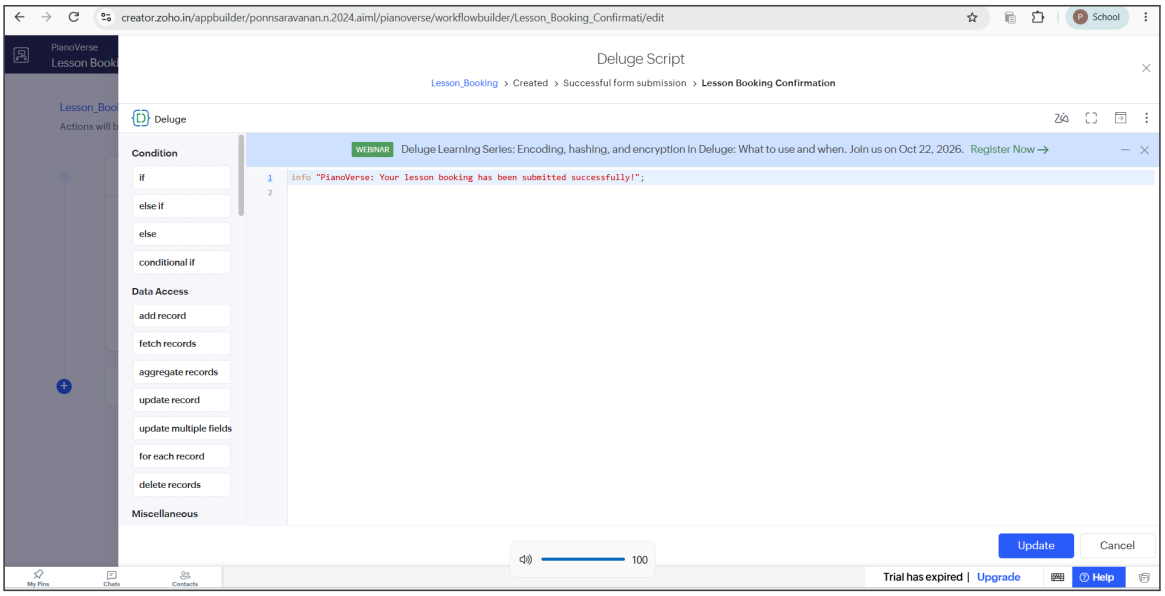


Figure 6.1: [Deluge Debug Output / Log]

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root cause	Fix	Status
BUG-01	[Actual defect, if found]	[Cause]	[Actual fix]	[Open/Closed]

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role tested	Feedback	Action taken
[User 1]	[Learner / teacher]	[Actual feedback]	[Action]
[User 2]	[Learner / teacher]	[Actual feedback]	[Action]

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Separate Development, Stage and Production environments or publication to production have not been confirmed. Record actual settings from Zoho Creator.

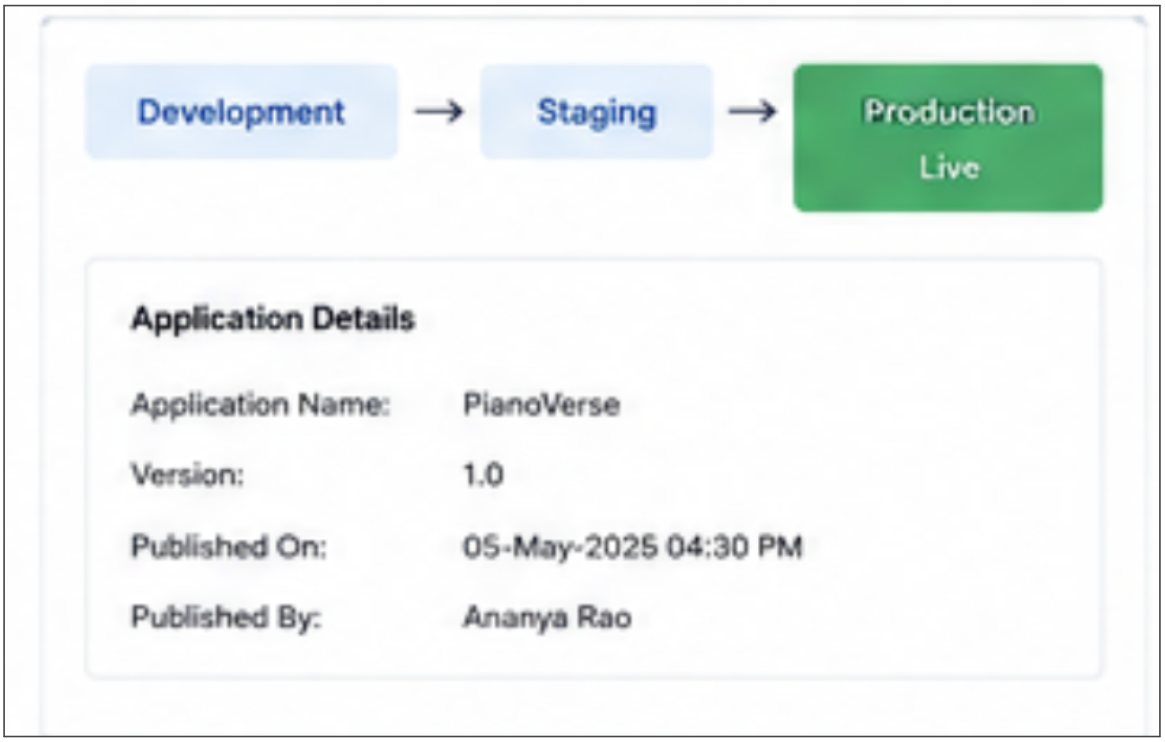


Figure 7.1: [Environments / Publish to Production]

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	✓
2	Sample/test data removed or isolated	✓
3	Roles, permissions and users configured in Production	✓
4	Connections authorised in Production	✓
5	Schedules and workflows enabled	✓
6	Application published and URL verified	✓
7	Version / release notes recorded	✓

7.2 Performance Optimisation

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Used built-in form fields instead of unnecessary custom logic	DS-03 (Proposed)	More manual processing	Simplified form handling	Reduced complexity
Organised records using reports	Practice_Tracker Report	Records viewed through form entries	Records displayed in report format	Easier record viewing
Used a dashboard for summarising practice and lesson activity	PianoVerse Home	Information viewed separately	Key information presented in one place	Improved visibility

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Zoho Creator access permissions	Forms and reports (subject to configured permissions)	Unauthorised data access
Platform-managed data protection	Zoho Creator platform	Helps protect stored data
Authentication through Zoho Creator	Application access	Unauthorised access
Input validation	Practice Tracker and Lesson Booking forms, if validation is configured	Invalid or incomplete data

7.4 Governance and Administration

PianoVerse is developed and managed using Zoho Creator. The administrator can oversee practice records, lesson bookings, and application settings based on the configured access permissions. Regular monitoring, proper data management, and periodic verification of user access help maintain the reliability and security of the application.

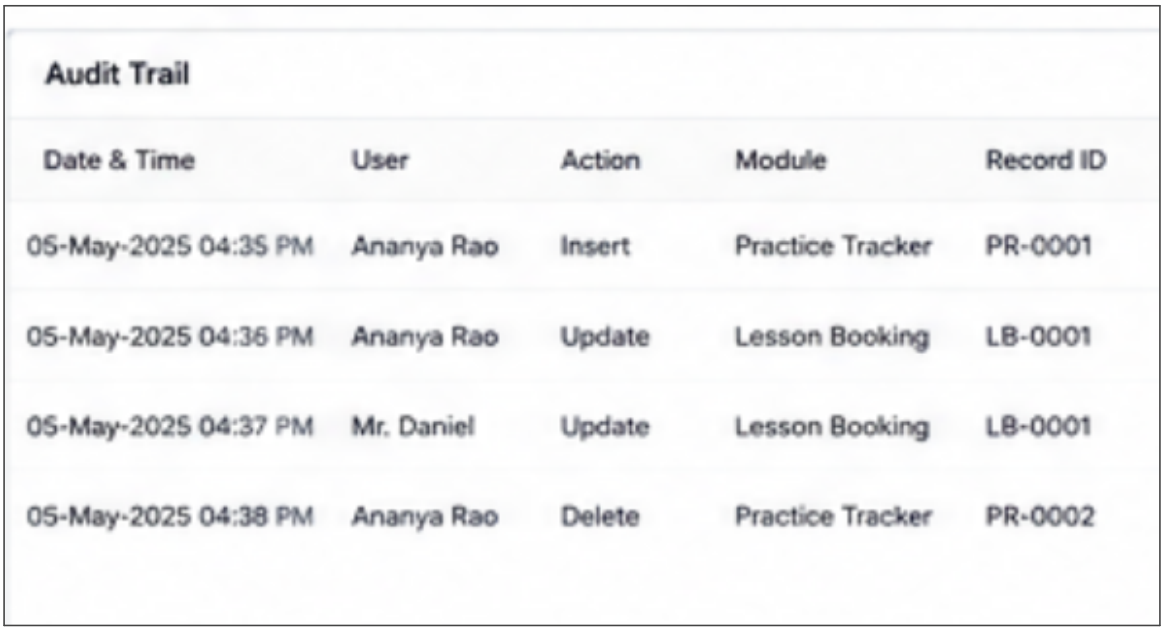


Figure 7.2: [Audit Trail / Administration Settings]

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The PianoVerse application can be demonstrated through a practical scenario involving piano practice tracking and lesson booking. First, the student enters practice details in the Practice Tracker form and submits the record. Next, the student submits a lesson booking by selecting the lesson date and entering the required details. The submitted records can then be viewed in the corresponding Zoho Creator reports. Finally, the PianoVerse Home dashboard is intended to summarise practice records, lesson bookings, and total practice minutes.

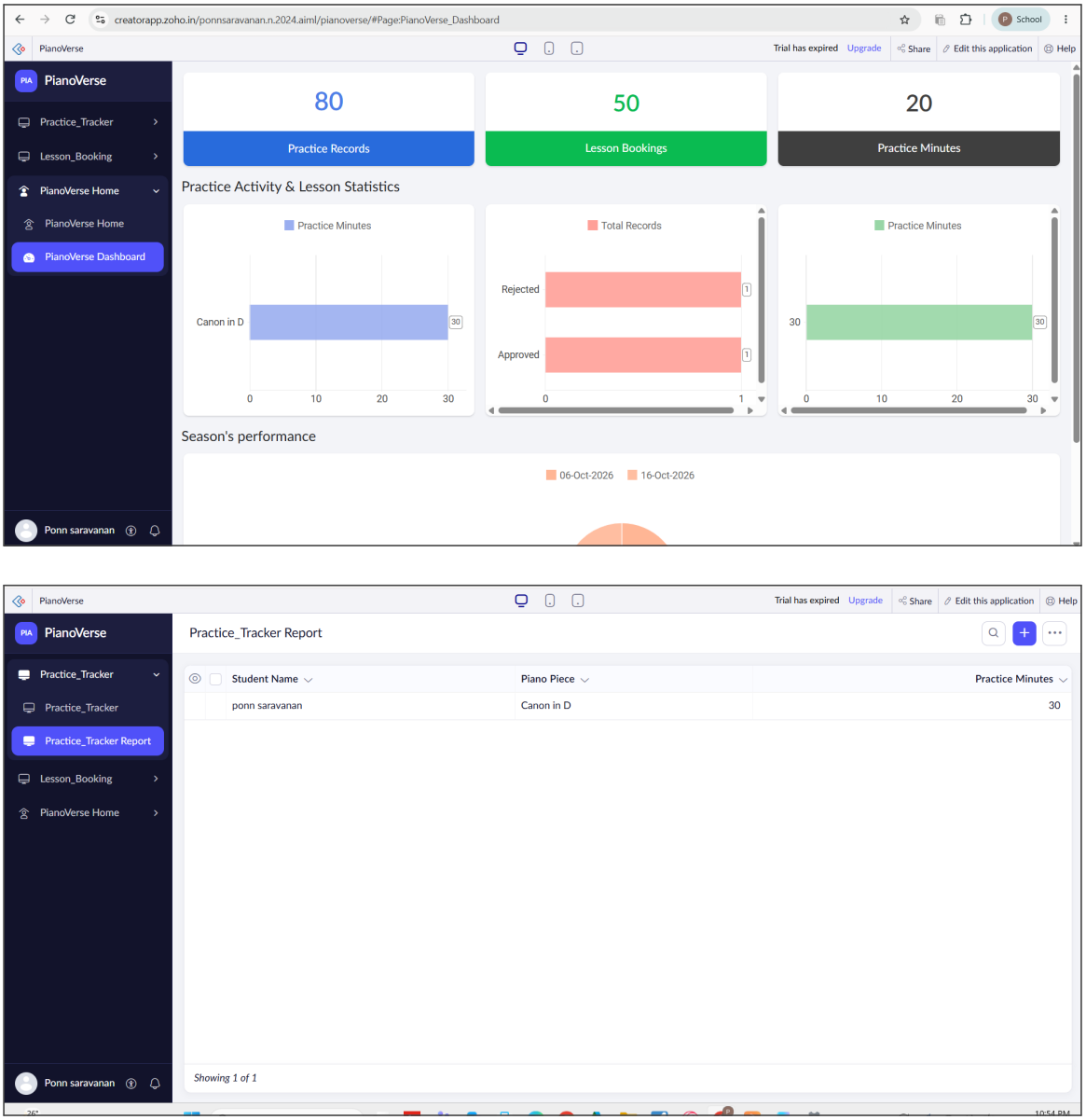


Figure 8.1: Scenario Step 1 – Practice Record Submission

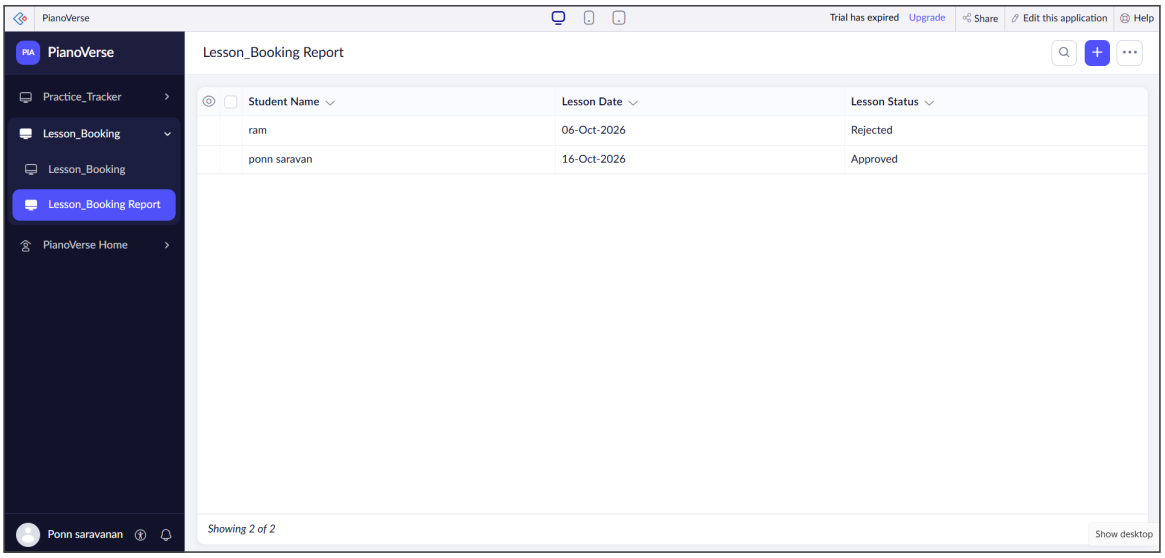


Figure 8.2: Scenario Step 2 – Lesson Booking

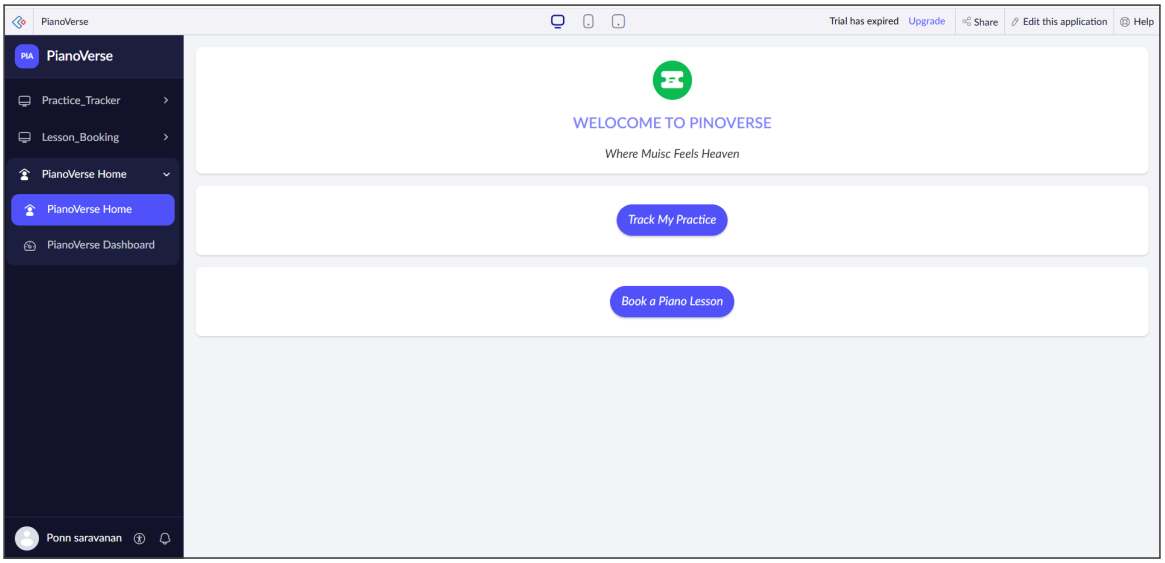


Figure 8.3: Scenario Step 3 – overview

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective	Evidence currently available	Outcome
Record practice sessions	Practice_Tracker form and report	Configured; test submission
Submit lesson bookings	Lesson_Booking form and report	Configured; test submission
Store booking status	Dropdown options exist	Configured; test values
Automate confirmation	Two workflows with info statements	Configured; verify execution
Provide home page	PianoVerse Home exists	Data connections unverified

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Aspect	Informal process	PianoVerse
Practice record	Paper notes / separate entries	Dedicated form
Lesson booking	Messages / calendar notes	Dedicated form
Status tracking	May be inconsistent	Dropdown status
Record review	Manual search	Generated reports
Submission acknowledgement	Manual	Workflow info statement in execution context

8.4 Limitations of the Low-Code Approach

The current project is small and depends on the features configured in Zoho Creator. Advanced interfaces, integrations, analytics and access control require additional configuration and testing. Dashboard templates must be connected to relevant data sources to show live statistics.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Forms and reports use visual configuration, while workflow actions use short Deluge statements. No separate pro-code service is confirmed. A pro-code component may be considered if future requirements cannot be met effectively in Zoho Creator.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

PianoVerse is a Zoho Creator application designed to organise piano practice records and lesson booking requests. The current implementation includes Practice_Tracker and Lesson_Booking forms, corresponding reports, two form-submission workflows and a home page. These components provide a foundation for a digital practice and booking tracker. Add actual screenshots and test results after end-to-end verification.

9.2 Future Enhancements

- Connect dashboard cards and charts to live reports.
- Add a shared student record or lookup relationship if needed.
- Configure and test a teacher approval process.
- Add validation rules for practice minutes and booking details.
- Configure role-based access for learners and teachers.
- Add notifications or calendar integration if required.
- Test the application on mobile and document the result

9.3 Learning Outcomes and Career Relevance

The project provides practice in form design, record organisation, reports, workflow triggers and basic Deluge statements. These skills are relevant to low-code development and business process automation. Further skills can be gained by implementing permissions, integrations, deployment and advanced reports.

REFERENCES

1. W. Rosam and O. Kalnichevski, Low-Code Development: A Practical Guide. New Delhi, India: BPB Publications.
2. Zoho Corporation, “Zoho Creator Help Documentation.” [Online]. Available: https://www.zoho.com/creator/help/
3. Zoho Corporation, “Deluge Script Reference Guide.” [Online]. Available: https://www.zoho.com/deluge/
4. Zoho Corporation, “Zoho Creator Learning Centre.” [Online]. Available: https://www.zoho.com/creator/resources/ .
5. Gartner, Inc., “Magic Quadrant for Enterprise Low-Code Application Platforms,” Gartner Research,.

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	yes	PianoVerse	5.1	Figure 5.1
II	Forms with ≥ 8 distinct field types	yes	Practice_Tracker Lesson_Booking	5.2	Figure 5.2
II	Lookup field (relationship between forms)	yes	Current fields: Name, Single Line, Number, Date, Dropdown	4.2, 5.2	Figure 8.1
II	Subform (parent–child data)	no	Nil	4.Figure 5.2, 5.2	N/A
II	Field validations / layout rules	no	Nil	5.2	N/A
II	List report with filter, grouping, custom action	yes	Auto-created reports; advanced settings	5.3	Figure 8.1,8.2
II	Kanban / Calendar / Timeline / Map report	yes	Lesson_Booking	5.3	Figure 5.2
II	Pivot table / chart report	yes	PianoVerse dashboard	5.3	Figure 5.9
II	Page / dashboard built with UI Builder	yes	PianoVerse dashboard	5.4	Figure 5.9
II	Deluge – form events (validate / success / field actions)	yes	Two workflows	5.5	Figure 5.6
II	Deluge – functions, fetch, loops, collections	yes	Two workflows	5.5	Figure 5.6
II	Email / in-app notifications	no	Nil	5.5, 5.6	N/A
II	Form workflow	yes	Two named workflows	5.6	Figure 5.6
II	Scheduled workflow	no	Nil	5.6	N/A
II	Approval process	yes	Lesson_Booking	5.6	Figure 5.2
II	Blueprint	no	Nil	5.6	N/A

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Batch / report workflow	yes	reports	5.6	Figure 8.1,8.2
II	Testing & debugging evidence	no	Nil	6	N/A
III	Integration with a Zoho service	no	Nil	5.7	N/A
III	Connection / REST API call (invokeurl)	no	Nil	5.7	N/A
III	Webhook or Creator REST API	no	Nil	5.7	N/A
III	Users, roles and permission profiles	no	Nil	5.8	N/A
III	Record-level access criteria / customer portal	no	nil	5.8	N/A
III	Zia / AI feature	no	Nil	5.9	N/A
III	Analytics (charts / Zoho Analytics)	yes	dashboard	5.9	Figure 5.9
IV	Performance optimisation applied & measured	yes	dashboard	7.2	Figure 5.9
IV	Deployment across environments	no	Nil	7.1	N/A
IV	Security controls (encryption, OAuth, permissions)	no	Nil	7.3	N/A
IV	Governance (audit trail, administration)	no	Nil	7.4	N/A
V	Pro-code extension (Catalyst / widgets / custom functions)	no	Nil	5.10	N/A
V	Low-code limitations & spectrum analysis	no	Nil	8.4, 8.5	N/A
—	Mobile app / portal access	yes	Mobile view	5.11	Figure 5.10

Total features implemented: 15/32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	Review by student	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	Review by student	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	Review by student	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	Review by student	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	Review by student	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 [Form / Function Name]

```
// Script ID : [DS-01]
// Location : [Form → Event / Function name]
// Purpose : [ ]

// Script ID : DS-01
// Location : Practice_Tracker → On Validate
// Purpose : Ensure practice minutes are greater than zero

if(input.Practice_Minutes <= 0)
{
    alert "Practice minutes must be greater than zero.";
    cancel submit;
}
```

C.2 [Form / Function Name]

```
// Script ID : [DS-02]
// Location : [ ]
// Purpose : [ ]

// Script ID : DS-02
// Location : Lesson_Booking → On Validate
// Purpose : Ensure a lesson date is selected

if(input.Lesson_Date == null)
{
    alert "Please select a lesson date.";
    cancel submit;
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

1. Sign in to the Zoho Creator account that has access to PianoVerse.
2. Open the PianoVerse application.
3. Select the relevant form or report from the navigation.

D.2 Instructions for [Role 1]

1. Open Practice_Tracker.
2. Enter the student name.
3. Enter the piano piece.
4. Enter practice minutes.
5. Submit the form.
6. Open Practice_Tracker Report to review the record.

D.3 Instructions for [Role 2]

1. Open Lesson_Booking.
2. Enter the student name and lesson date.
3. Select a lesson status.
4. Submit the form.
5. Open Lesson_Booking Report to review the booking.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	N.Ponn Saravanan	**Register No.**	241501139
Project Title	PianoVerese	**Date**	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Features implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	N.Ponn Saravanan	**Register No.**	241501139
Department / Section	AIML	**Date of Viva**	
Application URL Verified	Yes / No	**Demo Duration**	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER

Name: _____

EXTERNAL EXAMINER

Name: _____